```
import os

os.listdir()
```

```
['.config', 'univariable_analysis_practice.csv', 'sample_data']
```

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_csv("univariable_analysis_practice.csv")
```

```
df.sample(5)
```

	Age	Salary	Experience_Years	Department	City	Satisfaction
27	37	69000	10	Finance	Delhi	9
12	38	72000	12	Finance	Delhi	9
0	22	28000	0	IT	Jaipur	7
4	24	30000	1	HR	Jaipur	7
15	34	61000	9	IT	Jaipur	9

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 6 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Age               30 non-null     int64
 1   Salary            30 non-null     int64
 2   Experience_Years  30 non-null     int64
 3   Department        30 non-null     object
 4   City              30 non-null     object
 5   Satisfaction      30 non-null     int64
dtypes: int64(4), object(2)
memory usage: 1.5+ KB
```

```
df
```

	Age	Salary	Experience_Years	Department	City	Satisfaction
0	22	28000	0	IT	Jaipur	7
1	25	32000	1	HR	Delhi	8
2	28	41000	3	IT	Mumbai	6
3	31	52000	6	Finance	Pune	9
4	24	30000	1	HR	Jaipur	7
5	35	68000	10	IT	Delhi	8
6	29	45000	4	Marketing	Mumbai	5
7	27	39000	3	IT	Jaipur	8
8	40	85000	15	Finance	Pune	9
9	23	29000	0	HR	Delhi	6
10	32	56000	7	IT	Mumbai	8
11	26	35000	2	Marketing	Jaipur	7
12	38	72000	12	Finance	Delhi	9
13	30	48000	5	IT	Pune	8
14	21	26000	0	HR	Mumbai	6
15	34	61000	9	IT	Jaipur	9
16	28	43000	4	Marketing	Delhi	7
17	45	95000	18	Finance	Pune	10
18	36	70000	11	IT	Mumbai	8
19	25	33000	2	HR	Jaipur	7
20	29	46000	5	Marketing	Delhi	6
21	41	82000	14	Finance	Pune	9
22	33	59000	8	IT	Mumbai	8
23	27	37000	3	HR	Jaipur	7
24	39	76000	13	Finance	Delhi	9
25	24	31000	1	Marketing	Pune	6
26	31	51000	6	IT	Mumbai	8
27	37	69000	10	Finance	Delhi	9
28	26	36000	2	HR	Jaipur	7
29	43	88000	16	IT	Pune	10

Next steps: Generate code with `df` New interactive sheet

```
# 1.how big is the data
df.shape
```

```
(30, 6)
```

```
# 2.how does the data look like
df.head()
df.sample(4)
```

	Age	Salary	Experience_Years	Department	City	Satisfaction
0	22	28000	0	IT	Jaipur	7
6	29	45000	4	Marketing	Mumbai	5
23	27	37000	3	HR	Jaipur	7
7	27	39000	3	IT	Jaipur	8

```
# 3.what is the data type of columns
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 6 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
```

```
 0   Age              30 non-null     int64
 1   Salary           30 non-null     int64
 2   Experience_Years  30 non-null     int64
 3   Department       30 non-null     object
 4   City             30 non-null     object
 5   Satisfaction     30 non-null     int64
dtypes: int64(4), object(2)
memory usage: 1.5+ KB
```

```
# 4.are there any missing value
df.isnull().sum()
```

	0
Age	0
Salary	0
Experience_Years	0
Department	0
City	0
Satisfaction	0

dtype: int64

```
# 5.how does the data look mathematiclly
df.describe()
```

	Age	Salary	Experience_Years	Satisfaction
count	30.000000	30.000000	30.000000	30.000000
mean	30.966667	52100.000000	6.366667	7.700000
std	6.562607	20201.314399	5.281610	1.290549
min	21.000000	26000.000000	0.000000	5.000000
25%	26.000000	35250.000000	2.000000	7.000000
50%	29.500000	47000.000000	5.000000	8.000000
75%	35.750000	68750.000000	10.000000	9.000000
max	45.000000	95000.000000	18.000000	10.000000

```
# 6.are there duplicate values
df.duplicated().sum()#0 yaha row show kr raha hai ,agr koi or num hota to utni row duplicate hoti
```

```
np.int64(0)
```

```
# 7.how is the correlation b/w columns
df.corr(numeric_only=True)
```

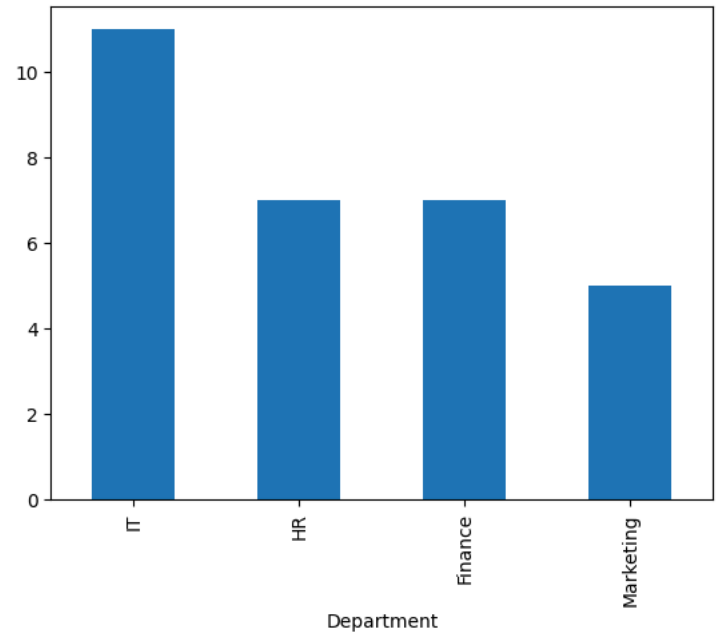
	Age	Salary	Experience_Years	Satisfaction
Age	1.000000	0.995181	0.993229	0.813072
Salary	0.995181	1.000000	0.997651	0.805367
Experience_Years	0.993229	0.997651	1.000000	0.810951
Satisfaction	0.813072	0.805367	0.810951	1.000000

```
#Categorical data analysis ke liye ab:



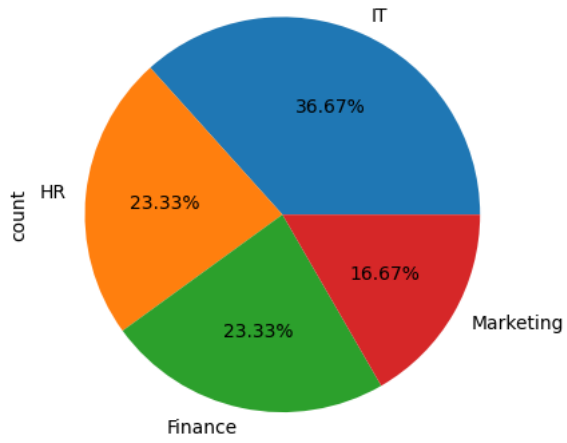
#value count ke liye
df['Department'].value_counts()
df['Department'].value_counts().plot(kind='bar')
# bars mai shoe ke liye
#sns.countplot(df['Department'])
```

```
<Axes: xlabel='Department'>
```




```
#% mai show ke liye pie chart
df['Department'].value_counts().plot(kind='pie',autopct='%.2f%%')
```

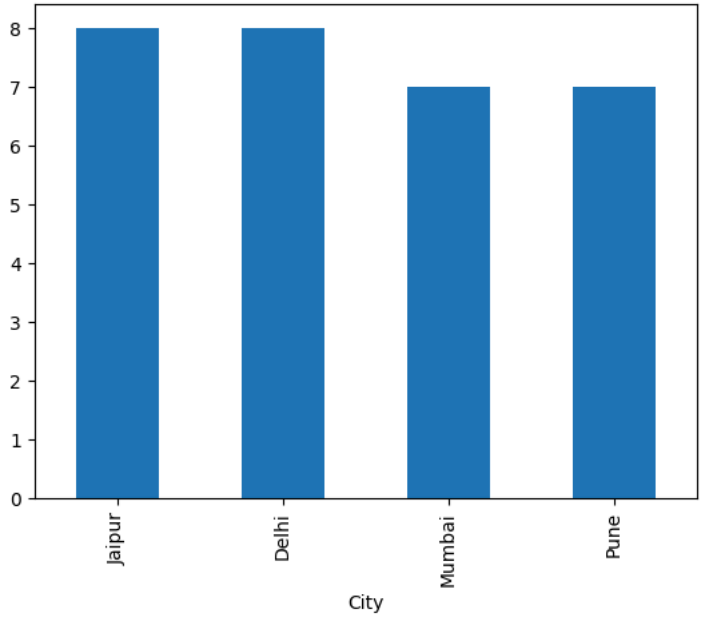
```
<Axes: ylabel='count'>
```




```
df['City'].value_counts().plot(kind='bar')
df['City'].value_counts()
#sns.countplot(df['City'])
```

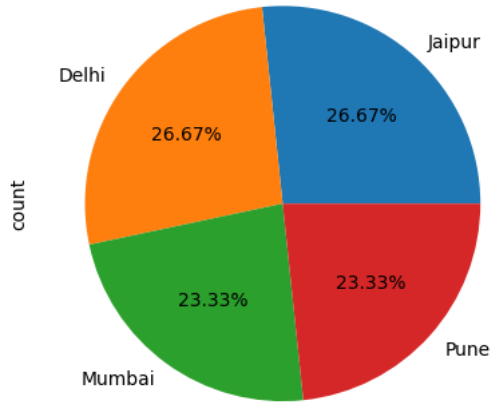
City	count
Jaipur	8
Delhi	8
Mumbai	7
Pune	7

dtype: int64




```
df['City'].value_counts().plot(kind='pie',autopct='%.2f%%')
```

```
<Axes: ylabel='count'>
```




```
df["Department"].unique()
df["City"].unique()
```

```
array(['Jaipur', 'Delhi', 'Mumbai', 'Pune'], dtype=object)
```

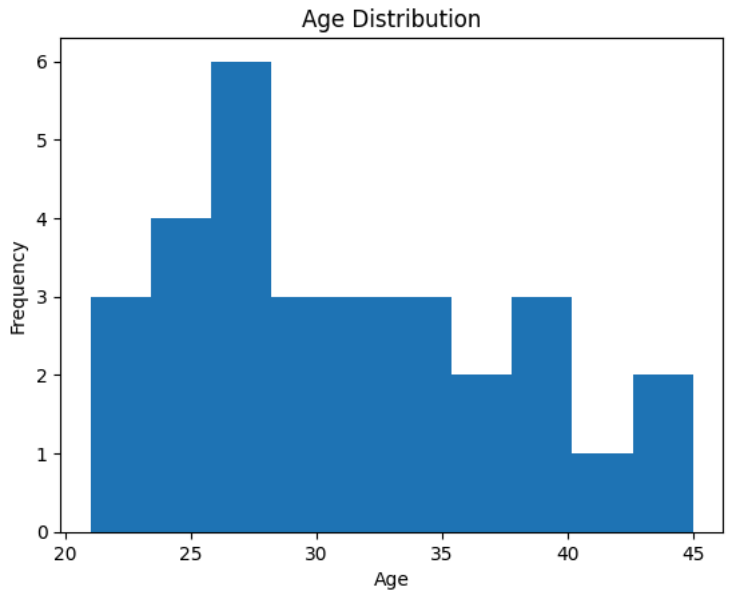
```
#numericl data  analysis ke liye ab:

df["Age"].describe()
df["Age"].mean()
df["Age"].median()
df["Age"].mode()
df["Age"].min()
df["Age"].max()
```

```
45
```

```
# histogram
```

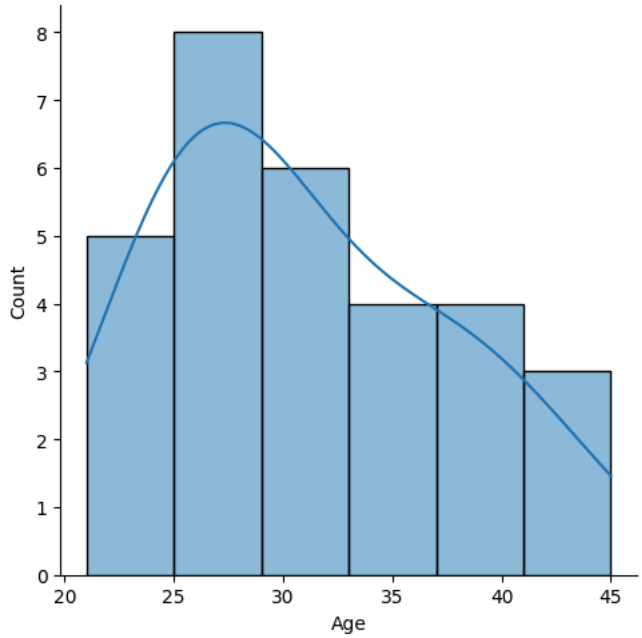
```
#plt.hist(df['Age'])
#plt.hist(df['Age'],bins=55)
plt.hist(df["Age"])
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.title("Age Distribution")
plt.show()
```




```
# distplot

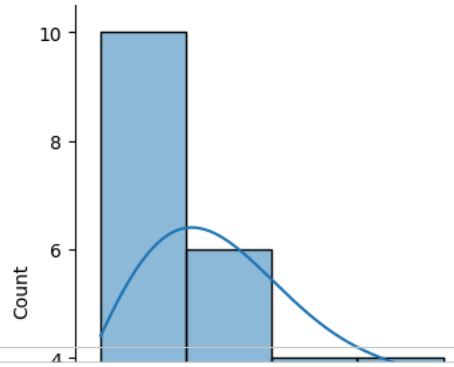
sns.displot(df['Age'],kde='true')
df['Age'].skew()
```

```
np.float64(0.49854572820209636)
```




```
sns.displot(df['Salary'],kde='true')
df['Salary'].skew()
```

```
np.float64(0.5725180122928911)
```




```
#boxplot


plt.boxplot(df["Age"])
plt.ylabel("Age")
plt.title("Age Boxplot")
plt.show()

sns.boxplot(df["Age"])
```

